

Requirement Analysis

Date	01 NOV 2025
Team ID	NM2025TMID05944
Project Name	Garage Management System
Maximum Marks	4 Marks

Data Flow Diagrams:

A **Data Flow Diagram (DFD)** is a visual representation of how data moves through a system. It illustrates the flow of information between processes, external entities, and data stores — helping understand how the system functions.

In the project “*To Supply Leftover Food to Poor,*” the DFD depicts how leftover food details are collected, processed, and delivered to needy people. The diagram shows interactions between **food donors, NGOs/volunteers,** and the **system database**. When donors post leftover food details, the system stores the data, alerts nearby NGOs or volunteers, and manages the pickup and delivery flow until the food reaches poor communities.

This helps visualize the workflow of donation requests, availability checks, volunteer assignments, and delivery confirmations, ensuring transparency and efficiency.

Example (Context-Level DFD):

- **External Entities:** Donor, Volunteer, NGO/Receiver
- **Processes:** Food Registration, Availability Notification, Volunteer Assignment, Delivery Confirmation
- **Data Stores:** Donor Database, Food Listing, Volunteer Records, Delivery Log

User Stories:

User stories describe how different users interact with the system to fulfill their roles in food collection and distribution. Each story focuses on a specific goal, acceptance criteria, and delivery priority.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Donor (Restaurant/Hotel)	Food Registration	USN-1	As a food donor, I want to register leftover food details so that it can be collected and distributed to	The system should allow donors to enter food type, quantity, and availability	High	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Volunteer	Food Collection & Delivery	USN-2	the poor. As a volunteer, I want to get notifications about available food near my area so that I can collect and deliver it on time.	time. The system should send alerts for available food within a defined radius.	High	Sprint-1
NGO/Coordinator	Verification	USN-3	As an NGO coordinator, I want to verify the food quality before delivery to ensure it is safe for consumption.	The system should record the verification and mark food as approved for delivery.	Medium	Sprint-2
System (Auto Process)	Notification	USN-4	As a system, I should automatically notify nearby volunteers and NGOs when new food donations are listed.	Notifications must reach all registered volunteers instantly after donor submission.	High	Sprint-2
Receiver (Needy People)	Feedback & Confirmation	USN-5	As a receiver, I want to provide feedback after receiving food so that service quality can be improved.	Feedback should be stored and viewable by admins for analysis.	Medium	Sprint-3

This structure ensures that every user' s needs are captured clearly — from donors contributing food, to volunteers collecting and distributing it, and NGOs verifying its quality. The DFD and user stories together establish the foundation for building a reliable, sustainable food distribution network.